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To cite this article: Koji Fujima, Kenji Masamura & Chiaki Goto (2002) Development of the 2D/3D Hybrid Model for Tsunami Numerical Simulation, Coastal Engineering Journal, 44:4, 373-397, DOI: [10.1142/S0578563402000615](https://doi.org/10.1142/S0578563402000615)

To link to this article: <https://doi.org/10.1142/S0578563402000615>



Published online: 10 Jan 2018.



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DEVELOPMENT OF THE 2D/3D HYBRID MODEL FOR TSUNAMI NUMERICAL SIMULATION

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Received 6 February 2002

Revised 30 August 2002

The 2D/3D hybrid tsunami numerical model was developed in which the conventional horizontally 2D model was adopted for the calculation in the wide region located far from the coastal structures while the 3D numerical model was used in the limited region adjacent to the structures. Applicability of the domain connection technique was examined by comparing the numerical results obtained by the present hybrid model with those obtained by applying the 3D model for the whole domain. Further, the results of the laboratory experiments were compared with the numerical results obtained by the hybrid model as well as the 2D model. It is shown that the present hybrid model reduces a calculation load significantly compared to the 3D model and is capable of reproducing the characteristics of three-dimensional and complicated flows around the structure, which cannot be reproduced by the 2D model alone.

Keywords: Tsunami numerical simulation; three-dimensional model; 2D/3D hybrid model; tsunami breakwater.

1. Introduction

In conventional numerical simulation of tsunami propagation, the horizontally two-dimensional (2D) model has been widely used on the basis of the finite difference method (e.g. Goto and Sato, 1993; Imamura, 1996). In the 2D model, the computation domain is divided into several sub-domains to employ the suitable basic equations and proper grid size. Typically, linear long-wave equations are solved with large grid in deep water, and nonlinear long-wave equations with small grid in shallow water. This method is applicable to a simulation for wide ocean area,

and is possible to reproduce the maximum run-up height accurately enough for practical purposes. Recently, Iwase *et al.* (1998) developed a practical model for tsunami propagation by using nonlinear dispersive long-wave equations solved with alternating-direction implicit (ADI) scheme. Their model provides a fairly accurate wave profile and run-up height distribution.

However, 2D models are based on the assumptions that the pressure distribution is almost hydrostatic and the horizontal velocity is almost uniform in a vertical direction. Thus, they cannot be applicable to tsunami behavior in a region around structures or a region where water depth varies abruptly. Accurate simulation of detailed velocity and pressure distributions around structures is necessary for the engineering purposes, e.g. evaluating the fluid force by a collision with a ship or drift, checking the stability of rubble mound foundation or submerged coastal structures, evaluating the bottom shear stress and so on.

A three-dimensional (3D) tsunami numerical model was developed by Masamura *et al.* (1996a, b) without the assumptions stated above. Masamura *et al.* (1996b) named this model the “non-hydrostatic pressure 3D model” to distinguish it from the multilayer model with hydrostatic-pressure assumption. Both full nonlinearity and full dispersion effects are taken into account in the non-hydrostatic pressure 3D model. However, in this model, the domain is divided into not only horizontal directions but also vertical direction, and the iterative computations are required. Thus, the computation time and the required memory become enormous in comparison to the 2D model. This was the major drawback for the non-hydrostatic pressure 3D model on its practical use.

In this paper, a practical 2D/3D hybrid simulation model for tsunami behavior around structures is developed, in which the non-hydrostatic pressure 3D model is adopted in the region where the three-dimensional fluid motion is predominant, and the 2D model is used in the wide region far from the structures.

In the 2D and 3D models, the computed variables are different because the discharge rate or depth-averaged velocity is computed in the 2D model, but the velocity with vertical distribution is in the 3D model. Thus, a proper domain connection technique is required to establish the hybrid model. To check the validity of the domain connection technique, several computations of the hybrid model were carried out by varying the area of the 3D domain, and the results were compared with those where the 3D model was applied for the whole domain.

In the previous non-hydrostatic pressure 3D model of Masamura *et al.* (1996b), the structure boundary and bottom topography were expressed by arranging the impermeable cells. Namely, a slope was approximated by composition of steps. Hence, the grid size is necessary to be set small to reduce the numerical error. This manner may cause the increase of computation cost. Instead, the present model adopts the porous-body method (e.g. Sakakiyama and Kajima, 1992; Isobe *et al.*, 2001). With this method, the complicated boundary can be modeled readily even if the grid size is not so small, hence the simulation is more efficient.

To examine the applicability of the present model, laboratory experiments were conducted for the tsunami behavior around tsunami-breakwater. The measured results were compared with the simulations by the hybrid model and the conventional 2D model.

2. Laboratory Model Experiments

First, an outline of the laboratory model tests is presented, because the same topography is used for the examination of the validity of the numerical method. The experiments were conducted by using the wave basin 7 m in width and 11 m in length as shown in Fig. 1. A piston type wave generator was installed at the left side of the tank. The other three sides of the tank were the vertical walls.

The tsunami-breakwater model having an opening was placed at the center of the wave tank. As shown in Fig. 2, the model consisted of a trapezoid mound foundation

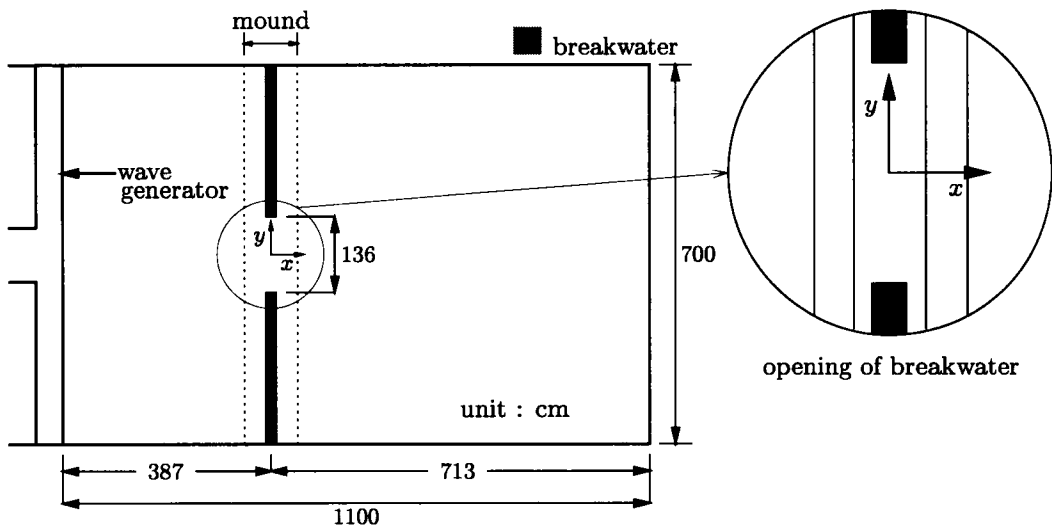


Fig. 1. Sketch of wave basin.

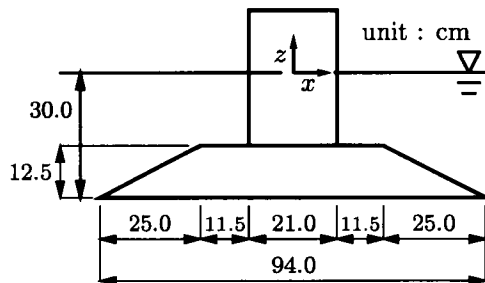


Fig. 2. Section of breakwater model.

94 cm in width and 12.5 cm in height and a rectangular caisson 21 cm in width. At the opening of the breakwater, the rectangular caisson model was not set on the mound foundation which was set even at the opening of the breakwater. This section is the scaled model of 1/200 of the prototype tsunami-breakwater in Suzaki bay, Kochi Prefecture, Japan.

In both experiment and numerical simulation, the origin of coordinates is set at the center of the breakwater opening at the still water level. The coordinate system is chosen as x being positive behind the breakwater (i.e. right side of Fig. 1), y being along the breakwater and z being positive upward. The region $x > 0$ is called "port side" and $x < 0$ "offshore side" in this paper.

The still water depth was 30 cm. From rest, the wave generator started to make a sinusoidal wave with a height of 2 cm and a period of 15 seconds (target value) at $t = 0$ s. The corresponding wave height and period in the prototype scale are 4 m and 210 s respectively. It is noted that the generated wave was not perfectly sinusoidal because the length between the wave paddle and the breakwater model was 3.87 m, much shorter than the wavelength of 25.7 metres.

Both water surface elevation and velocities were measured for 30 seconds from the start of wave generation. The time interval of data sampling was 0.05 seconds. The water surface elevation η was measured in the region $-80 \text{ cm} \leq x \leq 80 \text{ cm}$ and $-65 \text{ cm} \leq y \leq 0 \text{ cm}$ at 5 cm intervals by the capacitance-type wave gages. The velocities (u, w) were measured by the two-dimensional electromagnetic current meter, and the measurements were conducted on x - z planes at $y = 0, -20, -40, -65 \text{ cm}$, in the region $-80 \text{ cm} \leq x \leq 80 \text{ cm}$ and $-h \leq z \leq -2.5 \text{ cm}$ at 5 cm intervals,

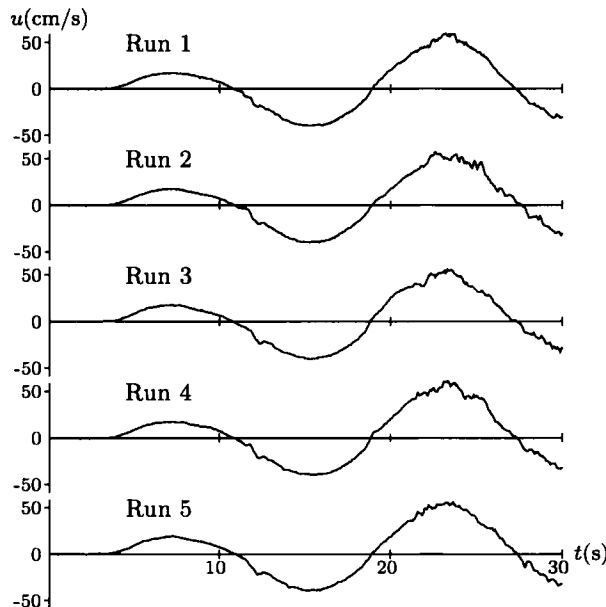


Fig. 3. Examples of measured data. (Velocity in x -direction at $x = -20 \text{ cm}$, $y = 0 \text{ cm}$, $z = -15 \text{ cm}$.)

where h was the local water depth. The time-history of wave paddle motion was also recorded, and all the measurements were synchronized in time.

In addition, the measurements of both water elevation and velocities were repeated at least 5 times for each point. Figure 3 shows the experimental data obtained by repeated measurements which demonstrate the repeatability of the experiments except for the small fluctuations. However, we found several cases in which the repeatability was not adequate. If the initial condition was not perfectly “still”, the current in the basin (remains of former measurements) might affect the behavior of the large vortex generated by the breakwater. Thus, when there was a discrepancy in data, additional measurements were carried out to examine whether it was caused by the experimental error or not. As a result, the number of measurements for one measurement point became 20 at the maximum. The final data were then evaluated by taking both the moving average in 0.5 second and the ensemble average in repeated measurements for a given measurement point.

3. Numerical Procedure

3.1. Three-dimensional domain

3.1.1. Governing equations

In the present model, the computational domain is divided into rectangular-solid cells, and the velocity components (u, v, w) are defined at the mid-points of the cell boundaries and the pressure p at the center of the cell as shown in Fig. 4. The porous-body method is used for the efficient expression of the bottom topography and structure configuration (Isobe *et al.*, 2001; Sakakiyama and Kajima, 1992). This method is applicable to handle the permeability of sand bed or rubble works as well as the bottom gradient in each cell. In the numerical simulations shown in this paper, a permeable layer does not exist; the porous-body method is used only for modeling

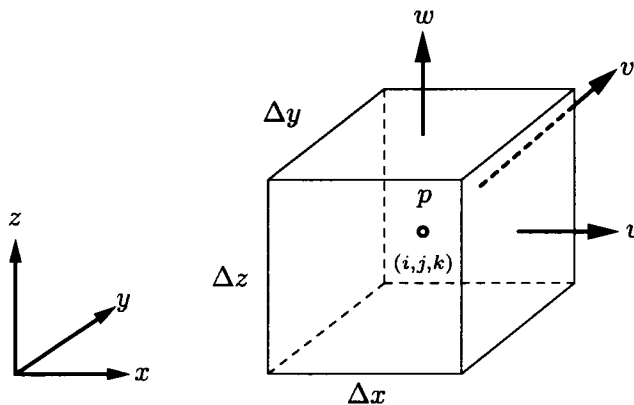


Fig. 4. Definition of variables for 3D model.

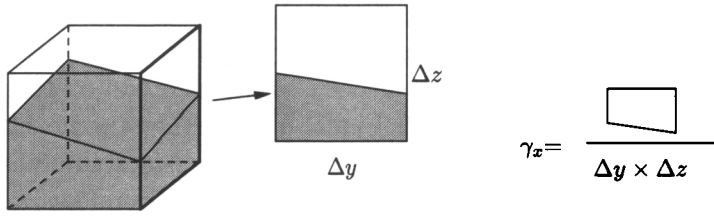


Fig. 5. Sketch of bottom cell. (Shaded area shows the impermeable bottom or structure.)

the bottom gradient in each cell. The governing equations in the 3D domain are as follows:

$$\gamma_v \frac{\partial u_i}{\partial t} + \frac{\partial \gamma_j u_i u_j}{\partial x_j} = -\frac{\gamma_v}{\rho} \frac{\partial p \delta_{ij}}{\partial x_j} + \gamma_v g_i + \frac{\partial \gamma_j \nu_t e_{ij}}{\partial x_j} \quad (1)$$

$$\frac{\partial \gamma_i u_i}{\partial x_i} = 0 \quad (2)$$

where x_i = coordinates (x, y, z); u_i = velocity components (u, v, w); ρ = density; p = pressure; g_i = gravitational acceleration; e_{ij} = strain rate ($\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}$); ν_t = eddy viscosity. The cell-surface permeability γ_i and the porosity γ_v are the parameters introduced to express the configuration of the bottom and structures in each cell. The parameter γ_i denotes the ratio of the permeable section area to the section area of the cell surface. For example, γ_x is determined by

$$\gamma_x = \frac{\text{section area through which fluid can pass}}{\Delta y \times \Delta z} \quad (3)$$

as shown in Fig. 5. The parameter γ_v denotes the ratio of the permeable volume to the total volume of the cell. Namely,

$$\gamma_v = \frac{\text{volume where fluid can occupy}}{\Delta x \times \Delta y \times \Delta z} \quad (4)$$

where Δx , Δy and Δz represent the grid size in each direction. If there is a slope in the cell, the pressure per unit section area at the cell surface is $p\gamma_i$; then, for example, the pressure gradient in x -direction becomes $-\frac{\partial}{\partial x}(p\gamma_x)$. On the other hand, the x -component of the reaction force from the slope is approximated as $p\frac{\partial \gamma_x}{\partial x}$. In addition, assuming that the slope is uniform in the cell, the average value of γ_x at the right and left cell-surfaces may coincide γ_v . The first term in R.H.S. of Eq. (1) approximates the summation of these effects

$$\left(-\frac{\partial}{\partial x}(p\gamma_x) + p\frac{\partial \gamma_x}{\partial x} = -\gamma_x \frac{\partial p}{\partial x} \approx -\gamma_v \frac{\partial p}{\partial x} \right)$$

thus, adopting the porous-body method implies considering the bottom gradient in the cell.

For an engineering application, the $k - \epsilon$ model is often used for the turbulence closure. However, the $k - \epsilon$ model requires the fine grid system. Because it is impractical for tsunami numerical simulation to use very fine grids, the zero-equation turbulence model is obliged to be adopted. Nakatsuji *et al.* (1992) reported that the sub-grid scale (SGS) model provided surprisingly accurate results even when a very large grid was used for the simulation of tidal current. Hence, the SGS model used by Nakatsuji *et al.* (1992) is adopted in the present simulation model as follows:

$$\nu_t = (c\Delta)^2 \sqrt{e_{ij}e_{ij}} \quad (5)$$

$$\Delta = (\Delta x \times \Delta y \times \Delta z)^{\frac{1}{3}} \quad (6)$$

where $c = 0.2$, which is the ordinary value for the SGS model. It is noted that the SGS model was originally used in the large-eddy simulation (LES), which requires smaller grid than the $k - \epsilon$ model. In this paper, the SGS model is adopted as a convenient zero-equation model apart from its physical basis.

The boundary conditions at the free surface are expressed as:

$$p = 0 \quad (7)$$

$$\frac{\partial u}{\partial z} = \frac{\partial v}{\partial z} = 0 \quad (8)$$

The vertical velocities out of free surface are obtained by the continuity equation.

3.1.2. Algorithm

The intervals between the velocity computation and the pressure and water surface elevation computation are $\Delta t/2$. Suppose that the velocity at $(n - \frac{1}{2})$ th step and the pressure and surface elevation at $(n - 1)$ th step are already determined in the whole domain. The values at the next step are evaluated by the following procedure:

- (a) Water surface elevation at n th step is computed.
- (b) The equation of motion is explicitly solved. At first, the velocity at $(n - \frac{1}{2})$ th step and the pressure at $(n - 1)$ th step are used for evaluating the advection, diffusion and pressure gradient terms. The solution, $\mathbf{v}^*$, is used as the approximate value of the velocity at $(n + \frac{1}{2})$ th step.
- (c) $\mathbf{v}^*$ obtained by the procedure (b) may not satisfy the continuity equation. Thus, the pressure field is modified to satisfy the continuity equation in all cells by iterative computation. The velocity $\mathbf{v}^*$ is also modified correspondingly.
- (d) The modified pressure is used for the pressure gradient term, and the velocity at n th step, $\mathbf{v}^n = (\mathbf{v}^* + \mathbf{v}^{n-1/2})/2$, is used for the advection and diffusion terms. Then the procedures (b) and (c) are repeated.
- (e) After the equations of motion and continuity are satisfied, the obtained velocity is fixed as the value at $(n + \frac{1}{2})$ th step and the pressure at n th step.

In the procedure (a), the water surface elevation is explicitly evaluated by the following equation obtained by integrating Eq. (2).

$$\frac{\partial \eta}{\partial t} + \frac{\partial}{\partial x} \int_{-h}^{\eta} \gamma_x u dz + \frac{\partial}{\partial y} \int_{-h}^{\eta} \gamma_y v dz = 0 \quad (9)$$

Here, η denotes the water surface elevation and h the still water depth.

When the equation of motion is solved in the procedures (b) and (d), the pressure gradient term is approximated by the central difference with the second order accuracy. The upwind difference with the second order accuracy is used to approximate the advection term to keep the simulations stable. The upwind difference with the first order accuracy is adopted, if the computation point is close to the boundary or structure, since the difference with the second order accuracy cannot be evaluated.

In the procedure (c), the pressure p is modified to make $\partial \gamma_i u_i / \partial x_i$ zero in the cell by

$$\Delta p_{i,j,k} = \frac{-1}{\frac{\partial D_{i,j,k}}{\partial p_{i,j,k}}} D_{i,j,k}, \quad D_{i,j,k} = \left(\frac{\partial \gamma_{\ell} u_{\ell}}{\partial x_{\ell}} \right)_{i,j,k} \quad (10)$$

Through this modification on p and the corresponding corrections on velocities, the continuity equation becomes satisfied. However, because the pressure in a neighboring cell is modified, continuity equation in the cell becomes unsatisfied again. Thus, the iterative computation is required to satisfy the continuity equation in all cells. The actual convergence criterion was set as $|D_{i,j,k}| < 5 \times 10^{-3}$.

In addition, the Crank-Nicholson scheme is adopted to keep the second order accuracy in the time advancing. Namely, when the momentum equation is re-computed in the procedure (d), the advection and diffusion terms are evaluated by the velocity at n th step, $\mathbf{v}^n = (\mathbf{v}^* + \mathbf{v}^{n-1/2})/2$. Through these procedures, finally the velocity at $(n + \frac{1}{2})$ th step satisfies

$$\begin{aligned} u_{i+\frac{1}{2},j,k}^{n+\frac{1}{2}} &= u_{i+\frac{1}{2},j,k}^{n-\frac{1}{2}} + \frac{\Delta t}{\rho} \frac{p_{i+1,j,k}^n - p_{i,j,k}^n}{\Delta x_{i+\frac{1}{2}}} + \frac{\Delta t}{\gamma_v} \left[- \left(\frac{\partial \gamma_x u u}{\partial x} + \frac{\partial \gamma_y u v}{\partial y} + \frac{\partial \gamma_z u w}{\partial z} \right)^n \right. \\ &\quad \left. + \left(\frac{\partial \gamma_x \nu_t e_{xx}}{\partial x} + \frac{\partial \gamma_y \nu_t e_{xy}}{\partial y} + \frac{\partial \gamma_z \nu_t e_{xz}}{\partial z} \right)^n \right]. \end{aligned} \quad (11)$$

Both $v_{i,j+\frac{1}{2},k}^{n+\frac{1}{2}}$ and $w_{i,j,k+\frac{1}{2}}^{n+\frac{1}{2}}$ satisfy the similar equations.

3.2. Two-dimensional domain

The following nonlinear long-wave equations are used in the 2D domain:

$$\frac{\partial \eta}{\partial t} + \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} = 0 \quad (12)$$

$$\frac{\partial M}{\partial t} + \frac{\partial}{\partial x} \left(\beta_{xx} \frac{M^2}{D} \right) + \frac{\partial}{\partial y} \left(\beta_{xy} \frac{MN}{D} \right) + gD \frac{\partial \eta}{\partial x} + \frac{f}{D^2} MQ = 0 \quad (13)$$

$$\frac{\partial N}{\partial t} + \frac{\partial}{\partial x} \left(\beta_{xy} \frac{MN}{D} \right) + \frac{\partial}{\partial y} \left(\beta_{yy} \frac{N^2}{D} \right) + gD \frac{\partial \eta}{\partial y} + \frac{f}{D^2} NQ = 0 \quad (14)$$

Here, D = total depth ($h + \eta$), (M, N) = discharge rate in (x, y) -direction, $Q = \sqrt{M^2 + N^2}$, f = coefficient of momentum loss, and β = momentum correction factor (see Sec. 3.3).

The momentum loss due to sea bottom friction is modeled with the coefficient f . If Manning's formula is adopted,

$$f = f_B = \frac{gn^2}{D^{1/3}} \quad (15)$$

where n is Manning's friction coefficient and $n = 0.015 \text{ m}^{-1/3} \text{ s}$ is used in the present computation which corresponds to the concrete basin floor. When the computational domain of nonlinear long-wave theory includes the breakwater, the momentum loss (the pressure force imbalance across the breakwater) due to sudden contraction and enlargement of the section, needs to be considered. Such a momentum loss can be expressed as a part of the coefficient f , i.e. $f = f_B + f_D$, where f_D is the expansion loss. Goto and Sato (1993) adopted $f_D = 0.5$ for the numerical simulation for Kamaishi bay-mouth tsunami-breakwater. However, in the present computations, the opening of breakwater is not included in the 2D domain, thus, f_D does not need to be considered.

Manning's friction law is adopted not only in the 2D domain but also in the 3D domain. Namely, at the lowest cell adjacent to the bottom in the 3D domain, the following equation is used:

$$\frac{\partial \gamma_z \nu_t e_{xz}}{\partial z} \approx \frac{1}{\Delta z} ([\gamma_z \nu_t e_{xz}]_t - fu \sqrt{u^2 + v^2 + w^2}) \quad (16)$$

Here, $[\]_t$ denotes the value at the top of the cell.

3.3. Domain connection

The 3D domain is set in the 2D domain as shown in Fig. 6. The discharge rate on the interface between the 2D and 3D domains is included in the 2D domain for this staggered grid system. In Fig. 6, the water surface elevations are evaluated at the locations denoted by the circles, and the discharge rates or velocities are the triangles, and the solid marks denote the 3D values, and the open marks the 2D values. For simplicity, the domain connection scheme at the left interface parallel to the y -axis is discussed in this section. The similar technique is used at the other interfaces.

The computation procedures and the data sharing between the 2D and 3D domains are stated below. Here, the discharge rate on the interface is expressed as M_I . Figure 7 illustrates the schematic sketch for the procedures in x - t domain.

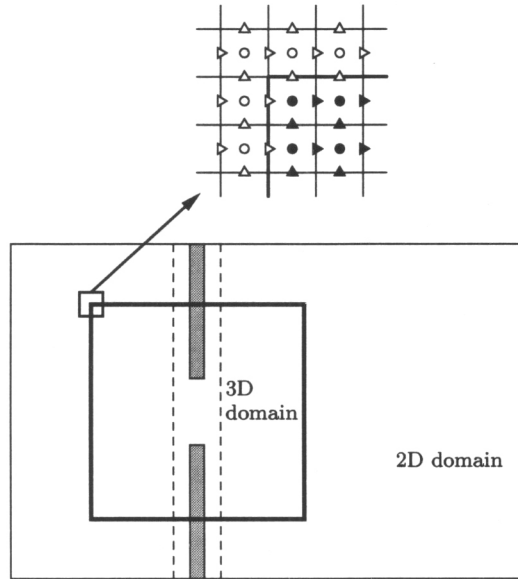


Fig. 6. Arrangement of 2D and 3D domains.

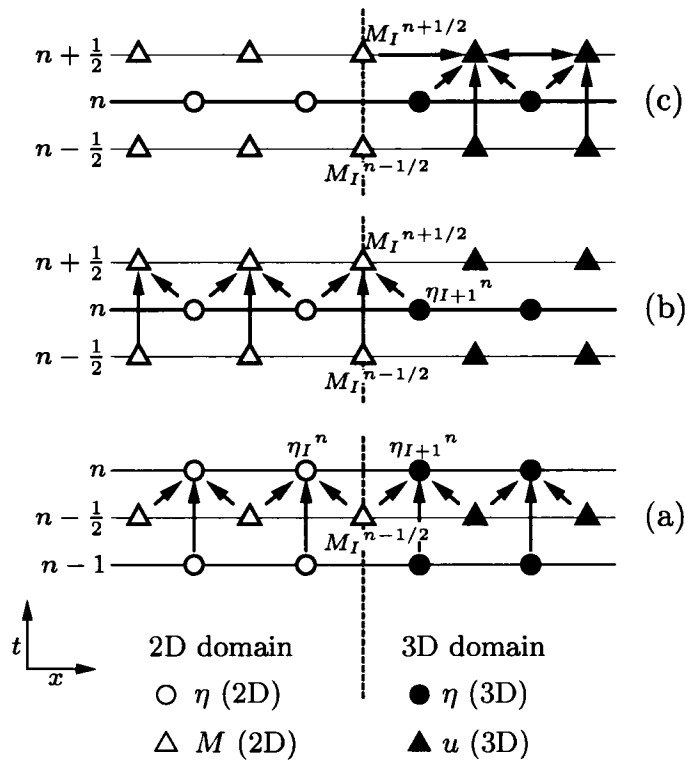


Fig. 7. Numerical procedures of domain connection.

- (a) Water surface elevation η is explicitly evaluated in both 2D and 3D domains. $M_I^{n-1/2}$ (included in 2D domain) is used to evaluate η_{I+1}^n (included in 3D domain).
- (b) Discharge rate in 2D domain is explicitly evaluated. The water surface elevation η_{I+1}^n is used to evaluate $M_I^{n+1/2}$. $M_{I+1}^{n-1/2}$ (included in 3D domain) is also used for the computation of $M_I^{n+1/2}$ to evaluate the advection terms in Eq. (13).
- (c) Velocity and pressure are implicitly computed in 3D domain. $M_I^{n+1/2}$ is used as the boundary condition for 3D computation.

At procedure (c), the 3D computation requires not only the discharge rate but also the vertical distribution of velocity at the 2D–3D interface. However, the 2D computation provides no information on the velocity distribution in the vertical plane. If the 3D domain is sufficiently large, horizontal velocity, (u, v) , at the interface of the 2D and 3D domains can be approximated by the long-wave theory. Thus, it is necessary to set the 3D domain sufficiently large so that the velocity at the interface can be approximated vertically uniform.

However, there may be a case where the 3D domain cannot be taken large enough because of the topography condition or restriction of computer memory. In such a case, a vortex with an axis parallel to x - y plane, which is called a “horizontal vortex” in this paper, possibly passes through the interface of the 2D and 3D domains. Then, the velocity distribution at the interface becomes non-uniform. If the uniform velocity distribution is used as the boundary condition under such situation, an inappropriate flow pattern is generated along the interface as shown in Fig. 8(2). This flow probably causes not only the decrease of accuracy but also the numerical instability. Thus, in order to hold the reasonable velocity distribution at the interface with sharing the information on discharge rate, $\frac{\partial u'}{\partial x} = 0$ is imposed to the boundary condition. By using this condition, an increase of w due to u' becomes zero, and then the flow along the interface is not generated. Here, $u = \bar{u} + u'$ and $\bar{u}$ is the depth-averaged velocity, u' the deviation from the averaged velocity. The averaged velocity $\bar{u}$ is determined by M_I , which is obtained by the 2D computation. In addition, both v and w out of the 3D domain are required for the 3D computation to evaluate the advection terms in the momentum equations in y - and z -directions. Then, the similar manner is used for v , and $\frac{\partial w}{\partial x} = 0$ is adopted for w . These boundary conditions allow that a horizontal vortex passes through the interface as shown in Fig. 8(3). No rational justification for this treatment can be declared but Fig. 8(3) clearly demonstrates the similar results as those where the 3D model is adopted in the whole domain.

Considering the velocity distribution at the interface, the total momentum transport across the boundary $\int_{-h}^{\eta} u^2 dz$ becomes larger than the momentum transferred by the averaged velocity M^2/D . In order to transfer the excess momentum transport due to horizontal vortex to the 2D domain, the momentum correction factor β is used. The parameters β_{xx} and β_{xy} appeared in the momentum equation in

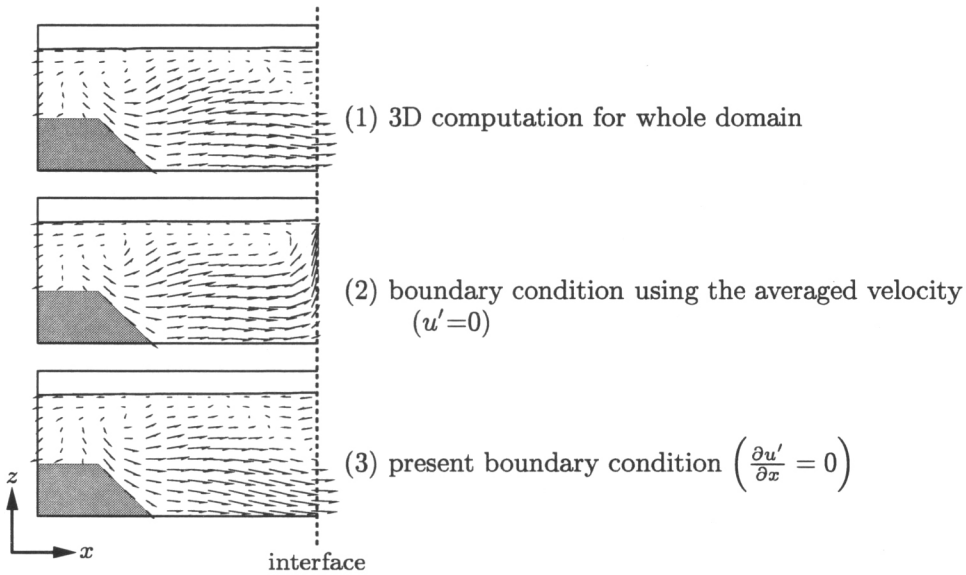


Fig. 8. Example of velocity vectors in x - z plane provided by several boundary conditions.

x -direction are written as follows:

$$\beta_{xx} = 1 + \frac{D}{M^2} \int_{-h}^{\eta} u'^2 dz \quad (17)$$

$$\beta_{xy} = 1 + \frac{D}{MN} \int_{-h}^{\eta} u'v' dz \quad (18)$$

If $\beta = 1$, the velocity distribution is uniform, and $\beta > 1$ implies the non-uniform velocity distribution. Thus, the effect of horizontal vortex crossing the boundary is transferred to the 2D domain by setting β greater than unity in the 2D domain. Ideally, the accurate value of β should be provided in the 2D domain to estimate the effect of horizontal vortex correctly. However, it is difficult to evaluate the accurate distribution of β in the 2D domain, because the nonlinear long-wave equation does not provide any information on horizontal vortex. In addition, setting a sharp distribution on β may cause a numerical problem. Thus, in the present model, the value of β on the interface is linearly reduced in the 2D domain so that β becomes unity at fifth grid from the interface. This treatment can be justified based on the numerical results where the 3D model was used for the whole domain. Actually, some numerical tests showed insignificant change even if the area of $\beta > 1$ was set to more than five grids.

By using β , it becomes possible to consider the momentum transport due to horizontal vortex from the 3D domain to the 2D domain. However, the above method cannot reproduce the situation such that the horizontal vortex does not dissipate in the 2D domain and is transferred back to the 3D domain again. Thus, there

is a limitation for setting the 3D domain too small. This will be discussed in the following section.

4. Comparisons of Numerical Results of the Hybrid Model and Those of the 3D Model for Whole Domain

Validity of the domain connection technique is examined in this section. If the hybrid model provides the similar results to those obtained by the simulation where the 3D model is used for the whole domain, then the domain connection technique appears to be valid.

Numerical simulations were conducted to reproduce the laboratory model experiment. By varying the area of the 3D domain as listed in Table 1, four cases of the simulations (Cases A through D) were conducted. Figure 9 shows the 3D domain for Cases A through D. In all cases, the 3D domain is the box-shaped area including the opening of breakwater, and the center of the 3D domain is set at the origin of

Table 1. Summary of grid size, grid number and computation time for hybrid model Cases A through D, 2D model and 3D model.

Model	Case	Grid Number $x \times y (\times z)$		Grid size (cm)	Computation
		2D	3D	$\Delta x \times \Delta y (\times \Delta z)$	Time (s)
Hybrid Model	Case A	220×140	$30 \times 30 \times 14$	$5 \times 5 \times 2.5$	900
	Case B	220×140	$50 \times 50 \times 14$	$5 \times 5 \times 2.5$	1700
	Case C	220×140	$90 \times 90 \times 14$	$5 \times 5 \times 2.5$	5600
	Case D	220×140	$130 \times 130 \times 14$	$5 \times 5 \times 2.5$	16000
2D model for whole domain		220×140	—	5×5	160
3D model for whole domain		—	$220 \times 140 \times 14$	$5 \times 5 \times 2.5$	27000

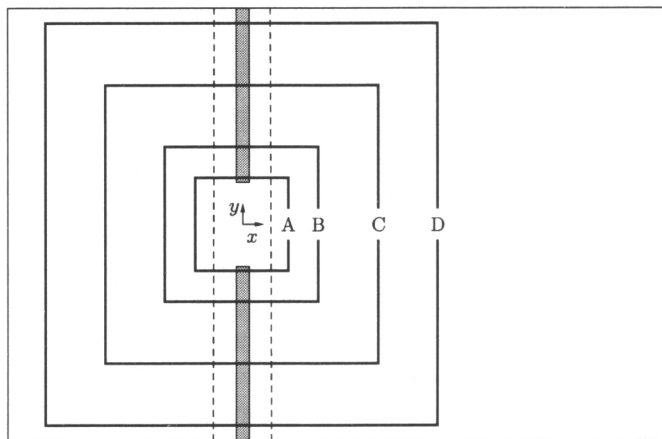


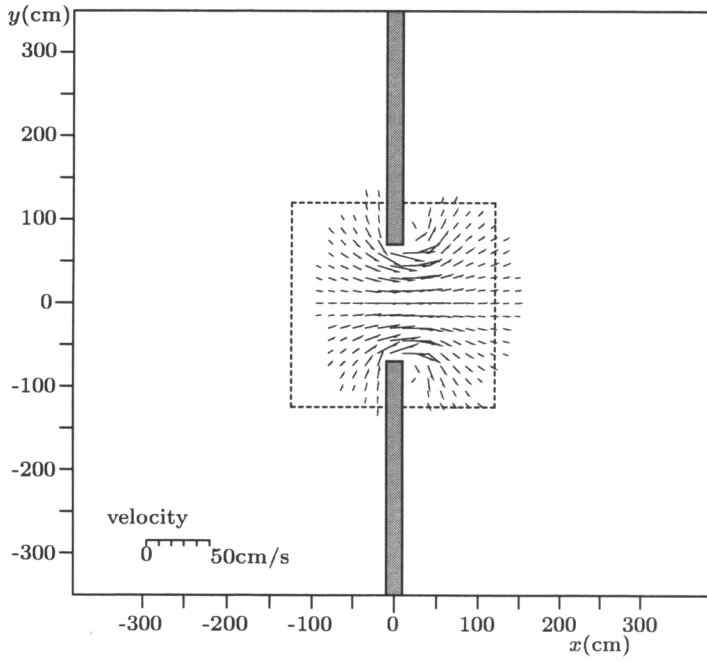
Fig. 9. 3D domain for hybrid model Cases A through D.

the coordinates. The length of the 3D domain in Case A is almost similar to the width of the opening, and that of Case D the width of the wave tank. The length of the 3D domain in Cases B and C is about two and three times the width of the opening respectively. Outside of the 3D domain is treated as the 2D domain which covers the entire wave tank. The interfaces between the 2D and 3D domains parallel to x -axis intersect the mound. Apart from the opening, three-dimensional effects of the flow should be negligible; namely, the velocity is uniform vertically even on the slope of the mound.

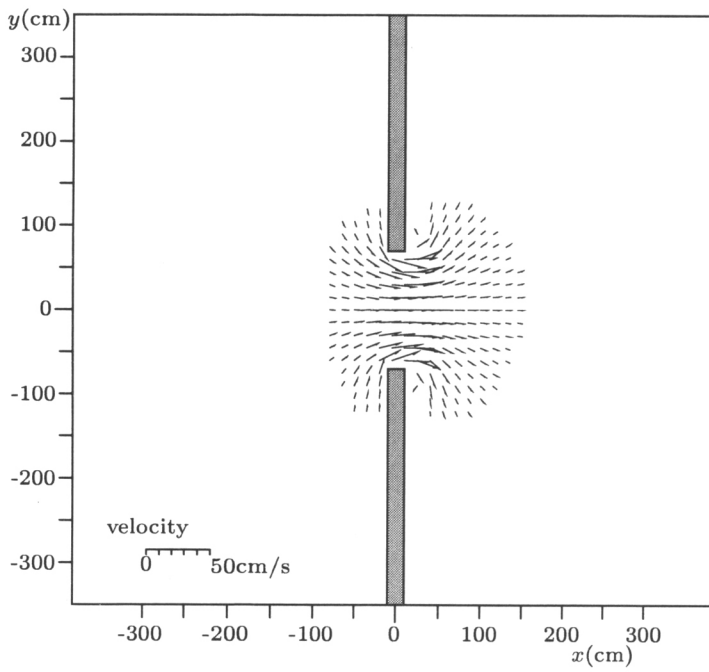
In Cases A through D, the grid number of the 2D domain is 220×140 , the grid size is $\Delta x = \Delta y = 5$ cm, and the time interval is $\Delta t = 0.01$ s. The discharge rate at the wave paddle, the edge of the 2D domain, is estimated by the paddle motion, i.e. the wave-generator boundary condition is used there. Because the other three sides of the tank are the vertical walls, the discharge rate across these walls are set at zero.

Spatial distributions of the depth-averaged velocity ($\bar{u}, \bar{v}$) obtained by the hybrid model (Case B) and the 3D model for the whole domain are compared in Figs. 10 and 11. Figures 10(a) and 11(a) show the results of the hybrid model, and Figs. 10(b) and 11(b) are those of the 3D model for the whole domain. The dotted lines in Figs. 10(a) and 11(a) indicate the interface of the 2D and 3D domains. A part of the whole domain, around the opening of the breakwater, is drawn in these figures; the right side of figures is the port side, and the left side the offshore side. In Fig. 10, the leading wave is passing through the opening of the breakwater, and the backflow of the leading wave appears in Fig. 11. Figures 10 and 11 prove that the hybrid model successfully reproduces the wave propagation passing through the interface of the 2D and 3D domains. Further, regarding the current around the opening, the hybrid model succeeds to produce the similar results as the simulation by the 3D model for the whole domain. Thus, it is concluded that the hybrid model provides the valid results for depth-averaged velocity.

Figure 12 shows the comparisons of velocity vectors in x - z section obtained by the hybrid model (Cases A through C) and the simulation of the 3D model for the whole domain. The result of Case D is not shown here because Case D results were quite similar to those of the 3D model. Only the 3D domains of the hybrid models are shown in Fig. 12, and for the 3D model for the whole domain, a portion of the computational domain is shown in the bottom figure of Fig. 12. The vertical section and the time in Fig. 12 were chosen so that the horizontal vortex appeared significantly. If the effect of vortex does not reach the interface, such as in Case C, the result of the hybrid model is in good agreement with that of the 3D model. In Case B, the horizontal vortex exists at the interface, and there is slight difference in the direction of velocity vector near the interface between the hybrid model and the 3D model, although the agreement is satisfactory in the location away from the interface. However, in Case A, the effect of horizontal vortex does not appear at the location where it should be. Case A cannot provide the accurate results for

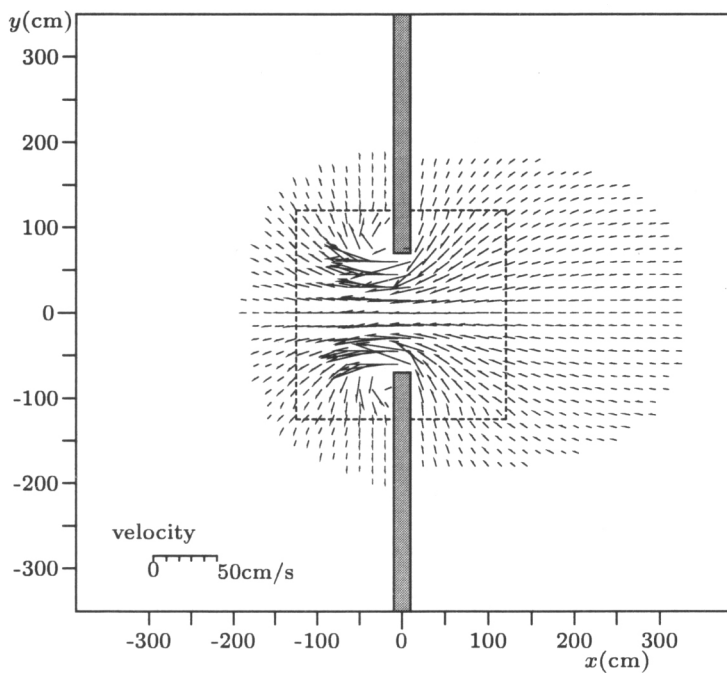


(a)

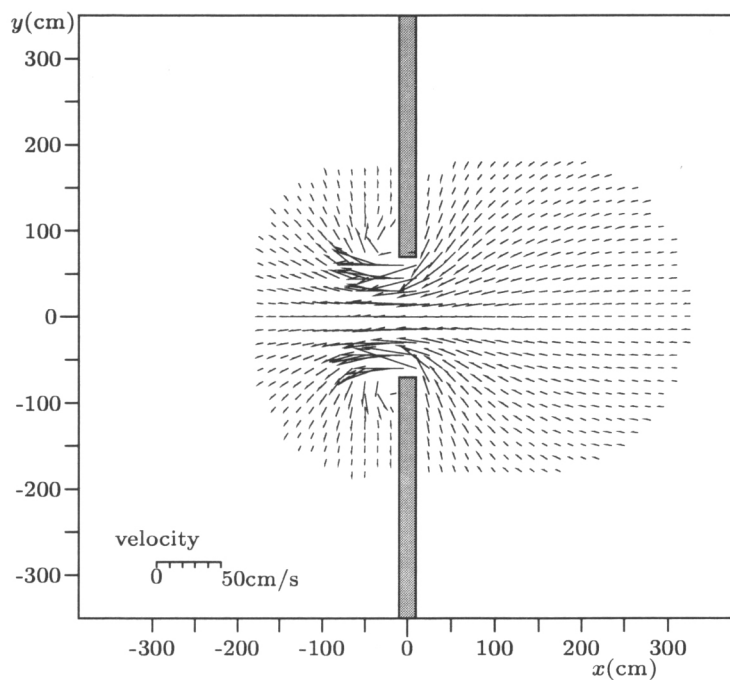


(b)

Fig. 10. Spatial distribution of depth-averaged velocity. ($t = 7$ s); (a) hybrid model Case B, (b) 3D model for whole domain.



(a)



(b)

Fig. 11. Spatial distribution of depth-averaged velocity. ($t = 14$ s); (a) hybrid model Case B, (b) 3D model for whole domain.

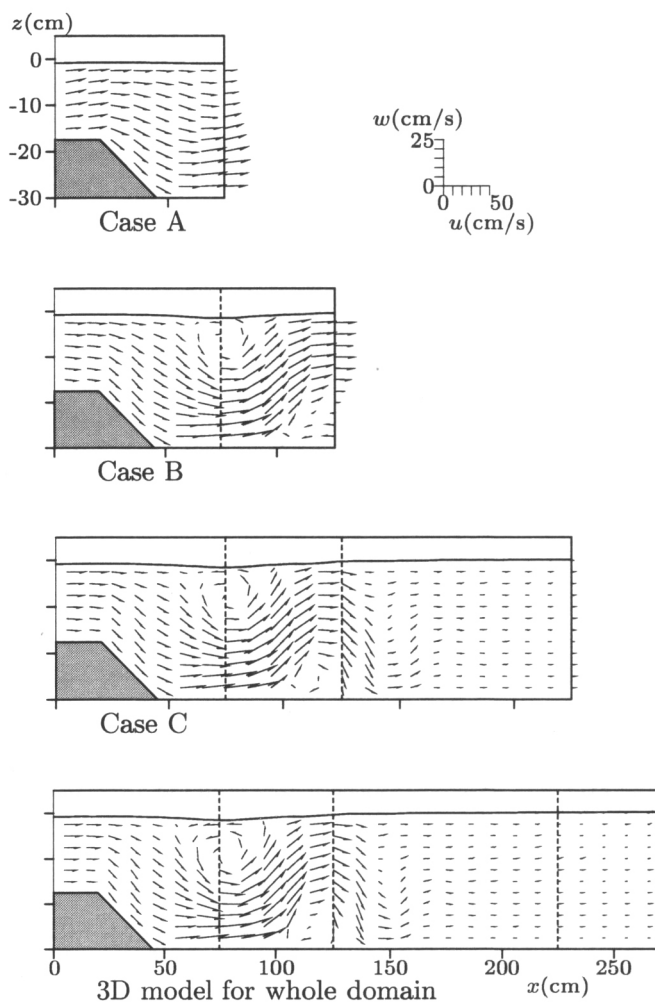


Fig. 12. Velocity vectors in vertical section $y = -40$ cm obtained by hybrid models and 3D model for whole domain. ($t = 26$ s.)

the vertical distribution of the velocity because the 3D domain is set too small. However, if the area of 3D domain is set as in Case B, the hybrid model is capable of producing the similar results to the simulation adopting the 3D model for the whole domain.

As a result, the connection technique between the 2D and 3D domains presented in this paper appears to be valid and makes the hybrid model simulation validated. When there is a horizontal vortex near the interface, slight errors are induced on the velocity distribution near the interface. However, the reproducibility of the flow away from the interface is found to be satisfactory. The hybrid model can provide sufficiently accurate information if its 3D domain is set to surround the region where three-dimensional flow is important.

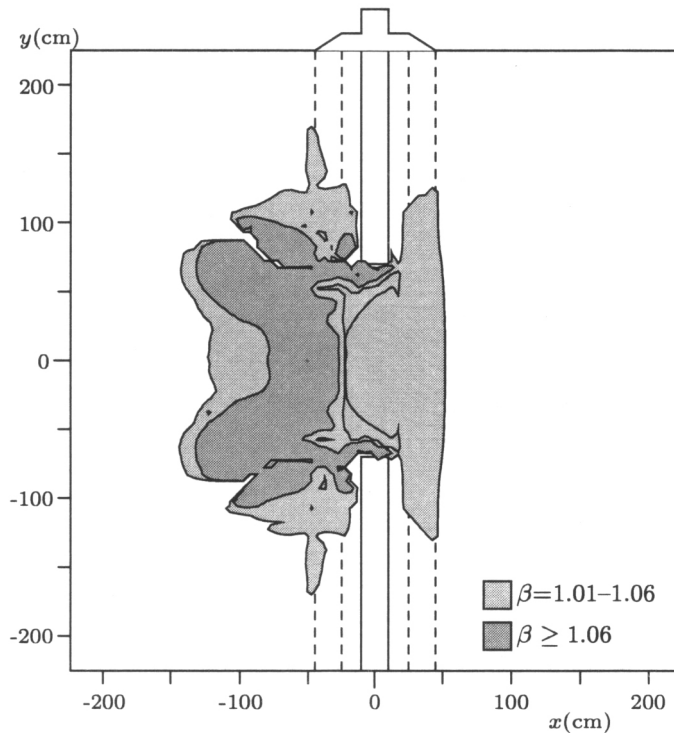


Fig. 13. Distribution of β_{xx} obtained by hybrid model Case C. ($t = 17$ s.)

Figure 13 shows the distribution of β_{xx} in the 3D domain in Case C, which is evaluated by the numerical results of the hybrid model Case C through Eq. (17). The contour lines of $\beta_{xx} = 1.01$ and 1.06 are drawn in the figure. This figure indicates that the influence of horizontal vortex reaches beyond the 3D domain of Case A. This also suggests that the 3D domain of Case A is too small and cannot provide the accurate results.

Required memory and computation time of the hybrid model should be also examined to discuss its advantages. Required memory is almost proportional to the grid number for the 3D domain; thus it can be saved by setting the 3D domain as small as possible. Computation time is shown in Table 1. It is clear that the hybrid model reduces the computation time significantly. Although Case A simulation has the problem on reproducibility, Case B provides the similar results as the simulation adopting the 3D domain for the whole domain. However, the computation time of Case B is only 1/16 of that of the simulation of the 3D model for the whole domain. In addition, Case D results are almost similar to Case C, although the computation time increases three times that of Case C. Therefore, the higher accuracy is achieved by setting the larger 3D domain, although excessively large 3D domain leads excessive computation cost. The size of the 3D domain should be determined by considering the objectives of the numerical simulation.

5. Comparisons of Experimental Results with Numerical Results Obtained by the Hybrid Model and Nonlinear Long-Wave Theory

In this section, to validate the present model, the hybrid model (Case B) results are compared with the experimental results (discussed in Sec. 2) as well as the conventional 2D (nonlinear long-wave theory) model results. In the conventional 2D simulation, $\beta = 1$ is used. Prior to the present discussions, several 2D computations were performed by varying the value of f_D . However, the computations with $f_D = 0$ appeared to yield the best results for the flow situation around the opening of breakwater. Thus, $f_D = 0$ is used for the comparisons in this section. Note that Goto and Sato (1993) obtained $f_D = 0.5$ as the best value of f_D through the discussions on the macroscopic momentum loss. Actual momentum loss is generated by the pressure force imbalance across the breakwater, which cannot be reproduced accurately by the nonlinear long-wave theory. Thus, the conventional tsunami numerical model introduces the additional (virtual) force on the opening of breakwater in the form of bottom friction to accord with the macroscopic momentum loss. However, this method possibly provides the improper effect on the local flow situation around the opening. This is the reason why $f_D = 0$ gave the best results in the preliminary examinations. This fact does not contradict Goto and Sato's result.

Figures 14 and 15 show the comparisons of the velocity vector (u, w) at the vertical sections $y = 0, -20, -40, -65$ cm around the opening of breakwater. The water surface elevation η is also compared in the figures. The left figures show the results of the hybrid model (Case B), and the right the experimental results. In Fig. 14(b), because the flow is sheltered by the breakwater mound, the vertical distributions of velocity become non-uniform behind the mound at the sections $y = 0, -20, -40$ cm. The conventional 2D simulation cannot reproduce this flow pattern, but the present model [Fig. 14(a)] reproduces the tendency of the experimental results. Regarding the flow pattern and surface elevation at $y = -65$ cm shown in Fig. 14, the computation results do not agree with the experimental ones. The disagreements are mainly induced by that the location of the vertical vortex generated at the edge of the opening of the breakwater cannot be simulated accurately, where the "vertical vortex" denotes the vortex with the vertical axis. The essential reason of this numerical error may be that the grid sizes were large in comparison to the scale of the vortex. However, for practical use, the grid size cannot be set so small. To improve the reproducibility without increasing the computation cost, the turbulence model should be re-examined. (Note that the disagreements shown on the top of the mound may be unavoidable because the location $y = -65$ cm is close to the edge of the breakwater opening $y = -68$ cm, while $\Delta x = \Delta y = 5$ cm.) In Fig. 15, the flow is separated into opposite directions at $x = -30$ cm in the section of $y = 0$ cm, and at $x = -50$ cm in $y = -20$ cm. This suggests that the flow due to the second wave is generated at the opening of the breakwater, while the effect of the vertical

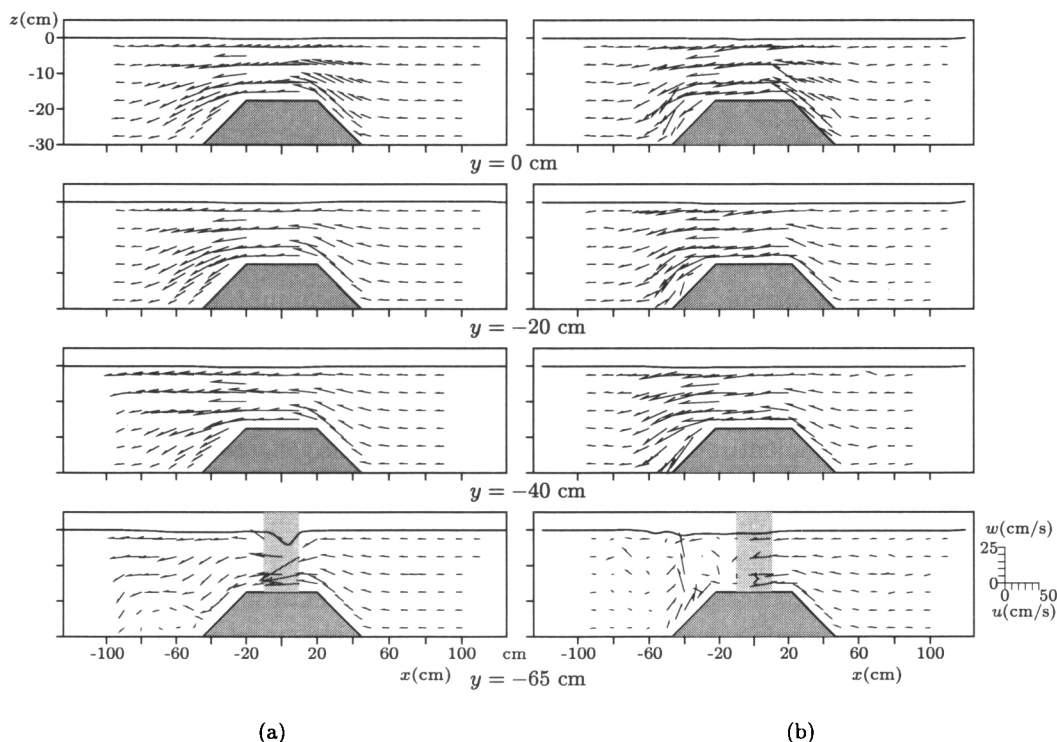


Fig. 14. Velocity vectors in vertical sections $y = 0, -20, -40, -65$ cm. ($t = 16$ s); (a) numerical results of hybrid model Case B, (b) experimental results.

vortex due to the backflow of the leading wave remains in the offshore side of the breakwater. In the section of $y = -65$ cm, there is the port-ward current with the magnitude of about 20 cm/s in the offshore side of the breakwater; and on the slope of the mound in the port side, the current blowing towards the offshore is generated. Further, on the horizontal bottom in the port side, the current becomes port-ward again. The present model can reproduce the complex flow characteristics as shown.

Figures 16(a)–(d) show the horizontal velocities u at the vertical sections $y = 0, -20, -40, -65$ cm around the opening of the breakwater. In each vertical plane, the vertical distributions of velocity measured at $x = -80, -60, \dots, 60, 80$ cm are plotted along the measured lines. The predicted velocities by the hybrid model are compared with those by the nonlinear long-wave theory as well as the experimental results. The interface of the 2D and 3D domains in Case B is at $x = \pm 125$ cm, although the results only in the region between $x = -80$ and 80 cm are shown in the figures because three-dimensionality is strong in this region. In Fig. 16(a), the velocity at $x = 0$ cm is the maximum and offshore-ward; in Fig. 16(c), the velocity at $x = 0$ cm is port-ward and the maximum. Figures 16(b) and (d) show the results of the intermediate phases. Hence Figs. 16(a) through (d) show the variation of the velocity field in about one wave period. The circles in these figures denote the experimental results, the solid lines the hybrid model (Case B), and the straight

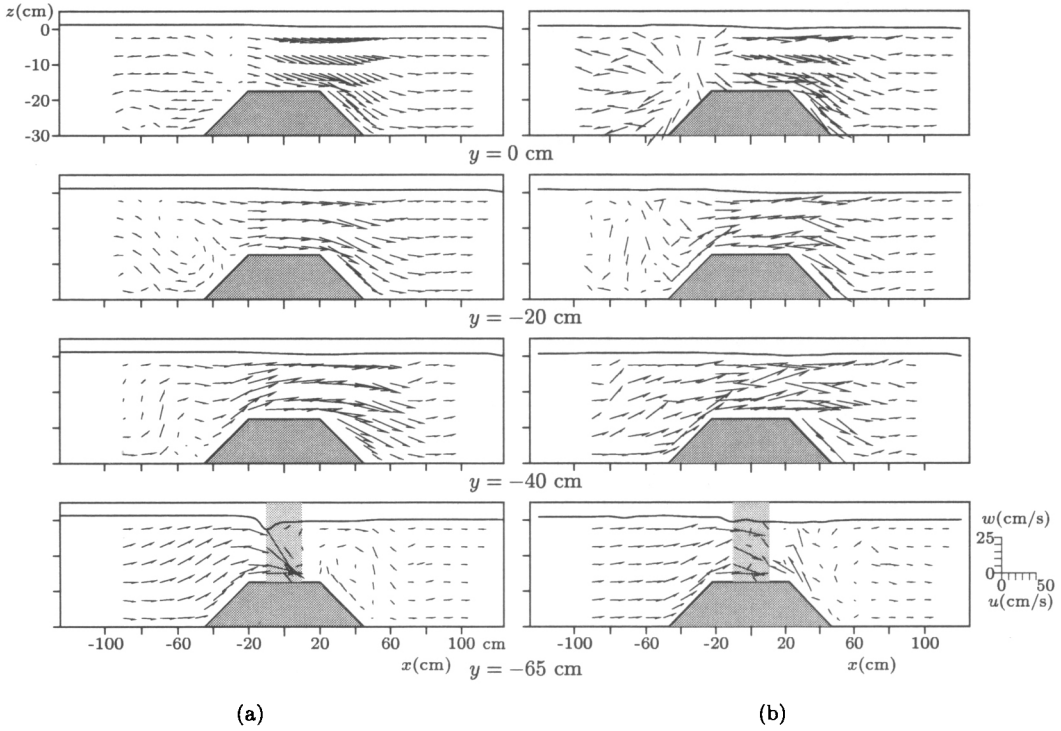


Fig. 15. Velocity vectors in vertical sections $y = 0, -20, -40, -65$ cm. ($t = 22$ s); (a) numerical results of hybrid model Case B, (b) experimental results.

broken lines the nonlinear long-wave theory.

At the locations where the velocity is vertically uniform, such as the entire port side in Figs. 16(a) and (b) or $x = 80$ cm in Fig. 16(c), both results of the hybrid model and the nonlinear long-wave theory agree with the experimental results accurately. At the opening of the breakwater ($x = 0$ cm), the nonlinear long wave theory also provides the adequate values except at $y = -65$ cm for all phases. It suggests that the conventional 2D numerical simulation can evaluate the inflow rate to the port accurately.

However, at several locations, such as $x = -40$ cm, $y = -20$ cm in Fig. 16(a) or $x = 40$ cm, $y = -40$ cm in Fig. 16(c), the surface velocity becomes large and the bottom velocity becomes small. The nonlinear long wave theory cannot provide the three-dimensionality in these locations, but the present hybrid model can reproduce the three-dimensional flow well.

In addition, at $x = 60, 80$ cm in the sections of $y = 0, -20$ cm in Fig. 16(d), the velocity distributions appear to be almost uniform, although the nonlinear long-wave theory underestimates the velocity considerably. At $y = -40$ cm at the same phase, there appears the adverse velocity near the water surface. This suggests that the vortex generated at the edge of the breakwater opening is advected there. Because of the existence of this vortex, the velocity at $y = 0, -20$ cm might increase due to

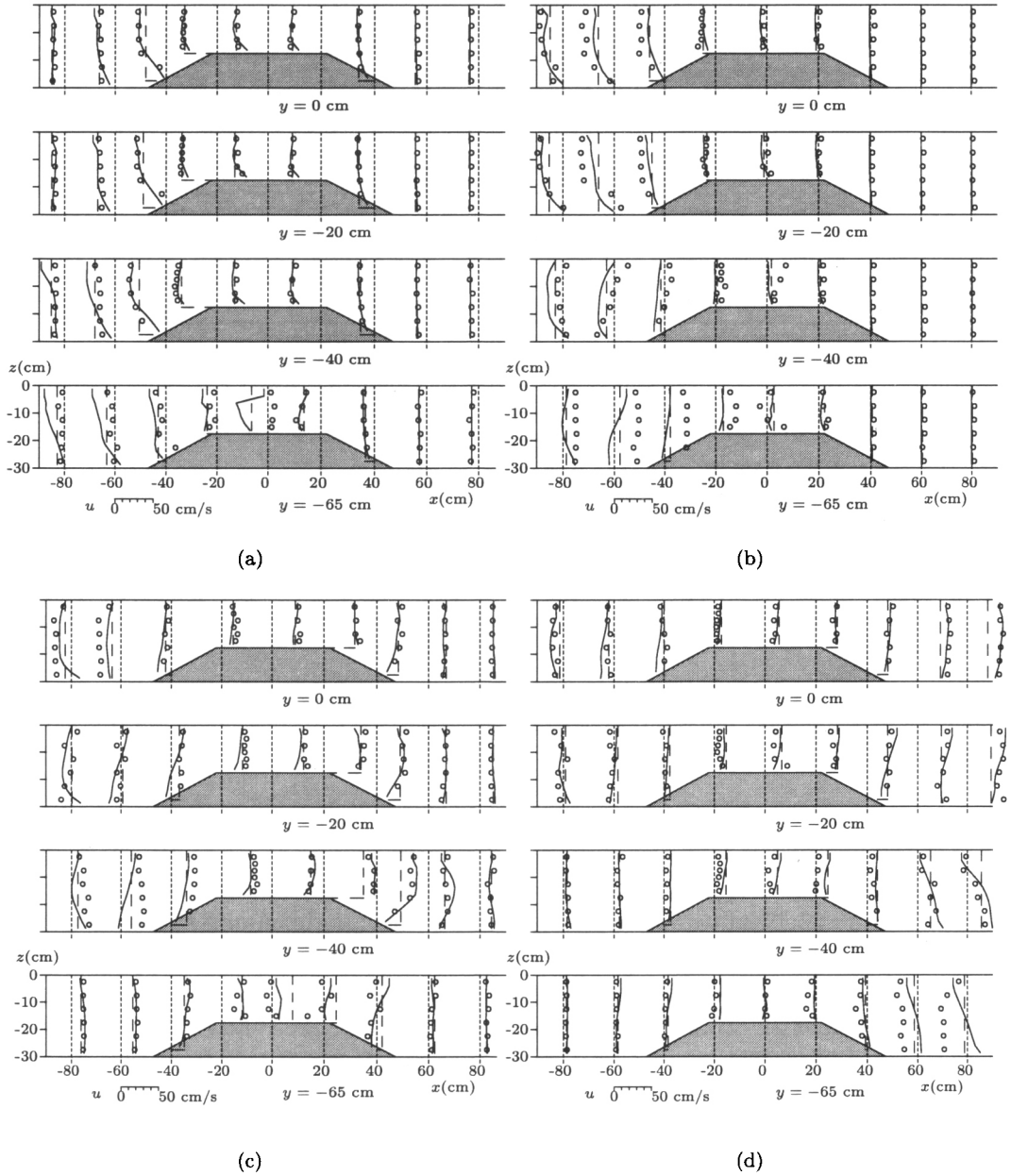


Fig. 16. Vertical distributions of horizontal velocity at $x = -80, -60, \dots, 60, 80$ cm in vertical sections $y = 0, -20, -40, -65$ cm. (○: experiment, —: hybrid model Case B, ---: nonlinear long-wave theory); (a) $t = 16$ s, (b) $t = 19$ s, (c) $t = 22$ s, (d) $t = 26$ s.

the flow contraction. The reason for underestimating the velocity at $x = 60, 80$ cm in $y = 0, -20$ cm might be that the nonlinear long-wave theory cannot reproduce the complicated behavior of the vortex. It suggests that the nonlinear long wave theory may become inaccurate even in terms of the averaged velocity when the flow three-dimensionality becomes strong. On the other hand, the present model appears

to be capable of reproducing such complicated phenomena. It is demonstrated that the hybrid model is useful for the numerical simulation for three-dimensional flows generated around structures.

However, at $y = -65$ cm, $x = -80, -60$ cm in Fig. 16(a), the measured velocities show approximately the uniform distribution, although the computed velocities have the curved distribution. In this region, both the horizontal vortex generated behind the mound and the vertical vortex generated behind the edge of the breakwater are interacted. In the experiments, the flow is well mixed by the vertical vortex and then the velocity distribution becomes uniform. However, in the simulations, the mixing effect is not reproduced sufficiently, then the computed velocity distribution becomes uniform 1 to 2 seconds after the experimental results. This induces the discrepancy appeared at $y = -65$ cm, $x = -80, -60$ cm in Fig. 16(a).

In all phases, the computed results obtained by the present model agree well with the experimental ones in the center region $|y| \leq 40$ cm, i.e. the main area of the flow passing the opening of the breakwater. From the engineering view for the tsunami disaster prevention, it is especially important to predict the variation of run-up height by the construction of tsunami breakwater, the stability of the submerged structure at the opening of breakwater, and the behaviors of floating bodies such as wood. For the accurate predictions for those, the prediction of the flows near the center region of the opening is most important. Thus, the present model is sufficiently accurate for the main engineering purposes. On the other hand, the present model cannot reproduce the behavior of the vortex generated by the edge of breakwater. In order to examine the stability of rubble mound at the edge of breakwater or the behavior of a ship caught in a vortex, the present hybrid model is inadequate. Improvement of the turbulence model may be needed, although the present model can be applied to the platform for the future model.

6. Conclusions

A new hybrid numerical simulation model for tsunami behavior around the structure was developed by combining the conventional 2D model and the non-hydrostatic pressure 3D model. In the developed technique for connecting the 2D and 3D domains, the discharge rate in the 2D domain is matched to that in the 3D domain. The water surface elevation, discharge rate and momentum correction factor in the 3D domain are also matched with those in the 2D domain. In the 3D domain, the velocity distribution at the interface to the 2D domain is determined to satisfy that the discharge rate agrees with the 2D value and that the velocity distribution inside the 3D domain keeps at the interface. By using this domain connection technique, the hybrid model provides the similar results as the case where the 3D model is used for the whole domain, if the 3D domain is set to be as in Case B, namely, approximately twice the width of the opening of breakwater.

In addition, the laboratory experiments were carried out and those results were

compared with the numerical results. The hybrid model satisfactorily simulated the complicated flow patterns on the mound foundation where the surface velocity deviated significantly from the bottom velocity. The required computation time of the hybrid model became much shorter than the case where the 3D model was used for the whole domain. Thus, the present model is efficient to examine the three-dimensional flow around a structure.

Because the developed hybrid model can accurately simulate the experimental flow fields in the main area around the opening of breakwater, it is sufficiently accurate for the practical numerical simulation from the view of tsunami disaster prevention. However, quick interaction of the horizontal and vertical vortices cannot be reproduced adequately, and the discrepancy between the numerical and experimental results becomes evident near the edge of breakwater. In order to improve the accuracy of the numerical simulations without increasing the computation cost, the turbulence model should be re-examined. However, even if the numerical treatment of turbulence will be improved, the basic concept of the developed model including the domain connection technique may be useful for the future model. In addition, the nonlinear long-wave theory was used as the governing equation in the 2D model in this paper, although any other long-wave equation, such as the nonlinear dispersive long-wave theory is also possible to be combined with the 3D model. If the dispersion effect is important in the outer area, for example, if the incident wave is a solitary wave, the nonlinear dispersive long-wave equation should be used in the 2D domain.

Acknowledgments

The authors would like to express their deep appreciation to Prof. Nobuo Shuto, Iwate Prefectural University, Japan, and Prof. Harry Yeh, University of Washington, USA, for their helpful comments on this paper. The authors also thank Prof. Toshiyuki Shigemura, National Defense Academy, Japan, and Prof. Kunihiko Iida, Tokai University, Japan, for their substantial discussions, and thank Satomi Naito and Aya Kosaka, former students of National Defense Academy, for their cooperation in conducting the experiments.

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